



Cohere Labs

Exploring the unknown, together



Data as Leverage:

Improving Foundation Models Beyond Scaling



Agenda

01

Motivation

02

Scaling
Plateau

03

Creating
Better Signal

04

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Better
Signal

05

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Takeaways

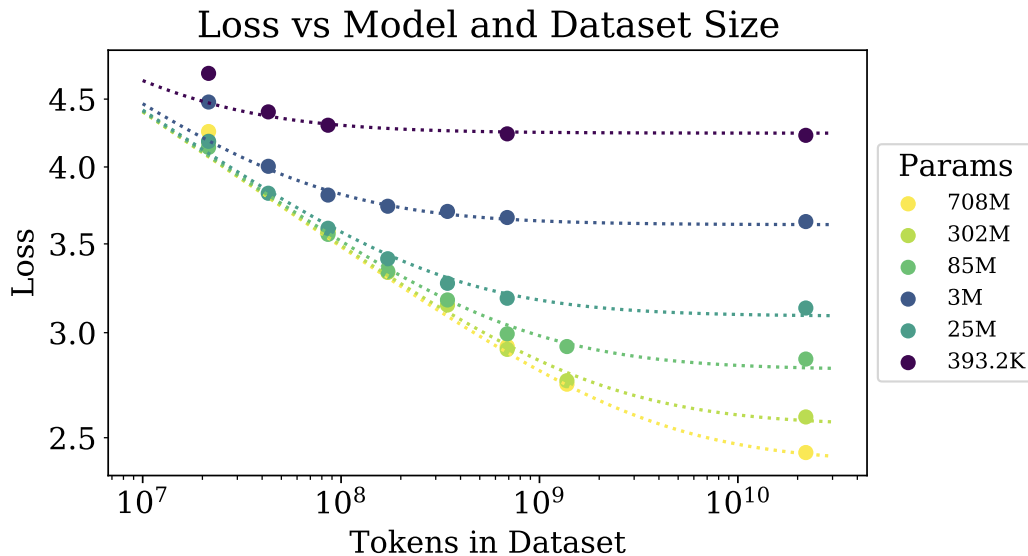
01

Motivation

Motivation

Trends that Scale

- High Frequency Use Cases
- High Resource Languages
- Broader Capabilities



Kaplan et al (<https://arxiv.org/abs/2001.08361>)

02

Scaling Plateau

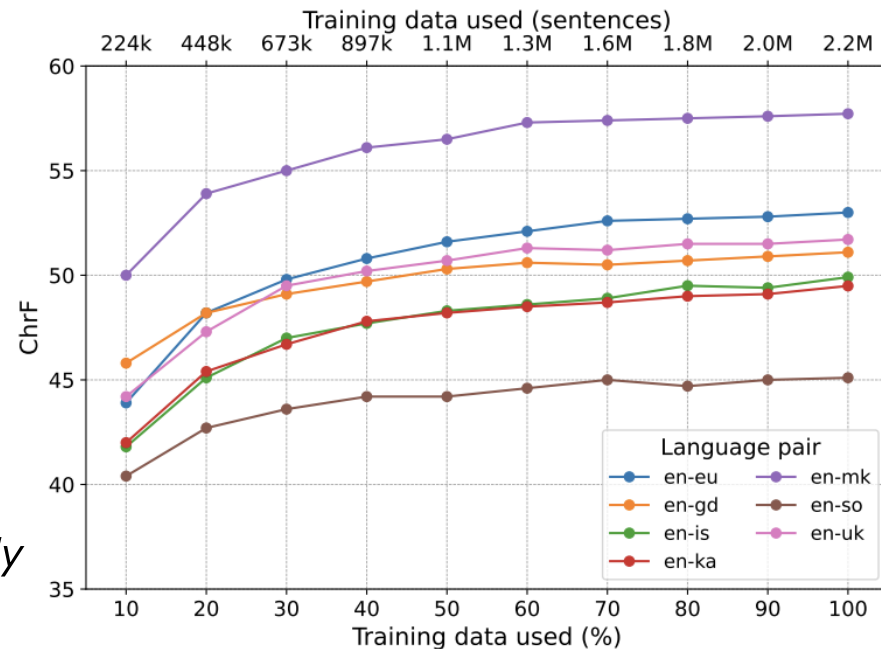
*Scaling shifts the curve upward,
but it doesn't change which curve you're on*

Where Scaling Plateaus

Is Scale the answer?

- Consistent improvements; trends upward
- Why does the “gap” remain consistent ?

If scale was enough, would these curves ideally converge ?

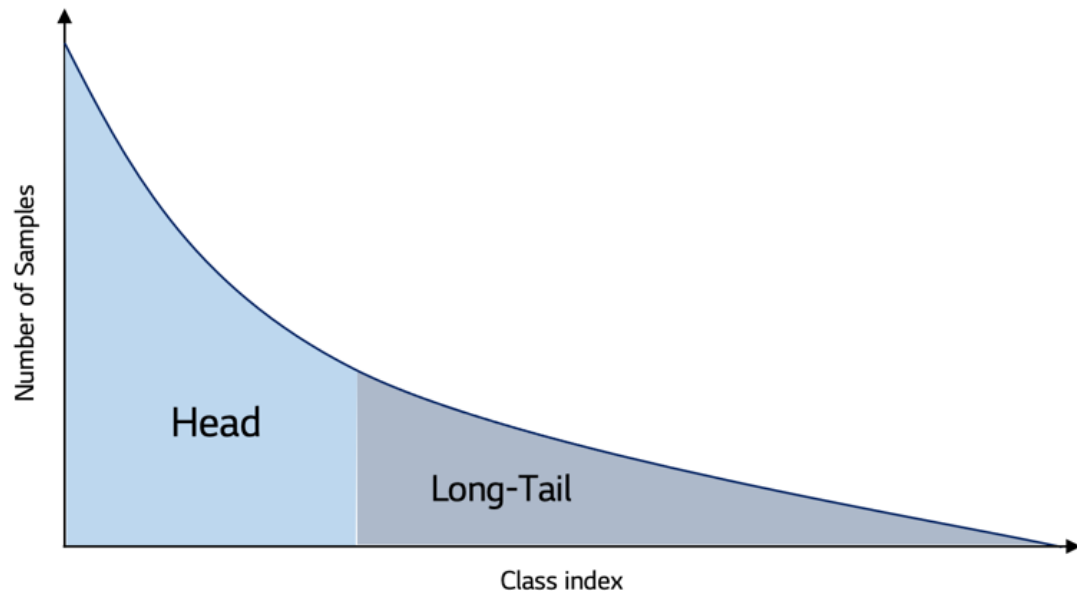


Gilbert et al
(<https://arxiv.org/abs/2505.14423v2>)

Where Scaling Plateaus

Trends that **don't** really scale well

- Low Frequency Use Cases
- Low Resource Languages
- Specific Targeted Capabilities



Can we do better?

Frameworks :

Leveraging Data

```
graph TD; A[Leveraging Data] --> B[Creating Better Signal]; A --> C[Revealing Better Signal];
```

Creating Better Signal

Multilingual Arbitrage
[EMNLP'25]

Ayomide Odumakinde, Daniel D'souza*,
Pat Verga, Beyza Ermis, Sara Hooker*

Revealing Better Signal

**Treasure Hunt: Real-time Targeting of the
Long Tail using Training-Time Markers**
[NeurIPS'25]

*Daniel D'souza, Julia Kreutzer, Adrien Morisot,
Ahmet Üstün, Sara Hooker*

03

Creating Better Signal

Why should we learn from just one ?

The Age of Synthetic Data

But is there a universal “best” multilingual model? 🤔

- 🤔 When was the last time your French Professor was your Chinese Professor?
- 🤔 When was the last time your Math Professor was your English Professor?

i.e *Why do we need to pick just one teacher ?* 🤔

The Age of Synthetic Data

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Multilingual Arbitrage

- There is no universal “best” multilingual model!
- Synthetic Data amplifies weakness i.e *mode collapse*
- Teacher Quality can vary by language, skill, task complexity, etc

i.e *Why not learn about specialized skills from specialized teachers?*

Multilingual Arbitrage

Setup

- Treat teachers as a portfolio.
- Route prompts to the best teacher available.
- Optimize data, not models

FROM

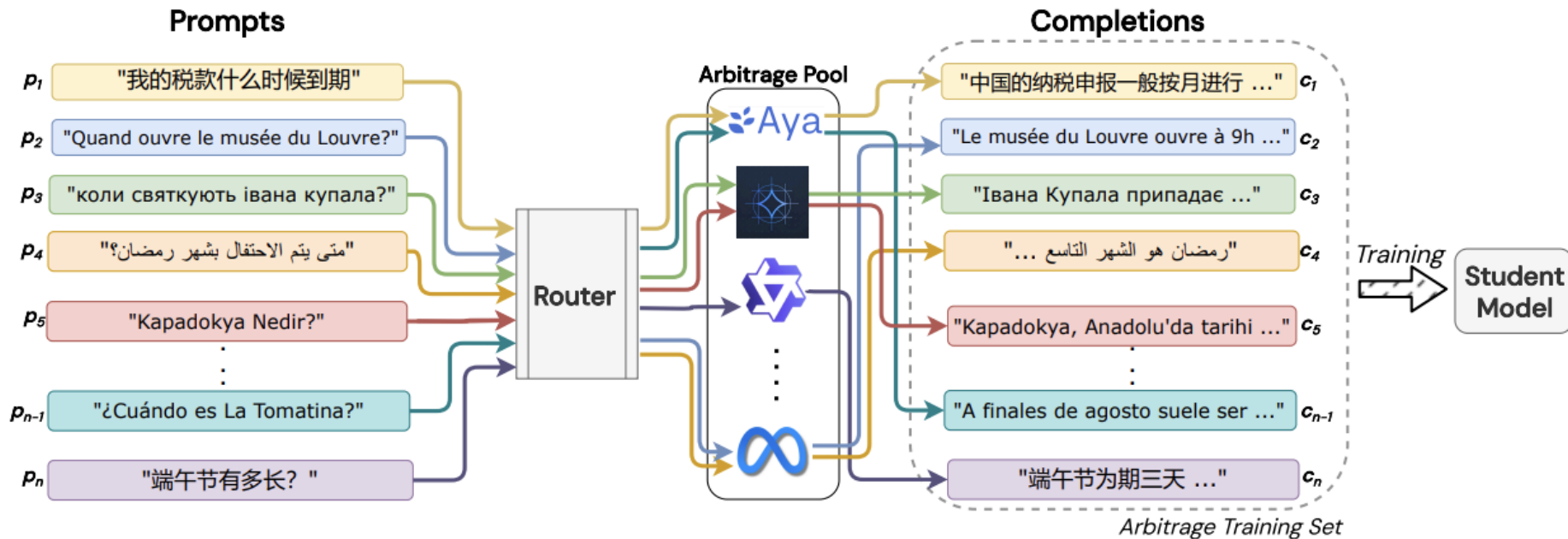
“Which model is best overall ?”

TO

“Which model is best for this prompt? “

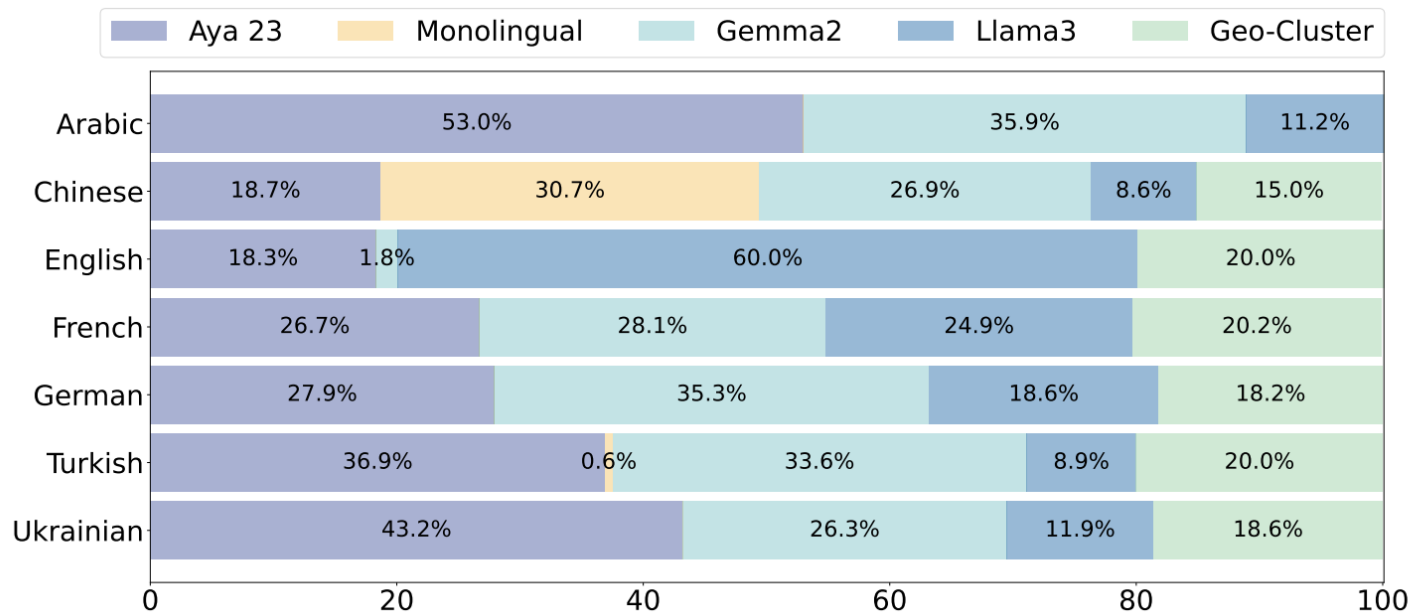
Multilingual Arbitrage

How does Routing work ?



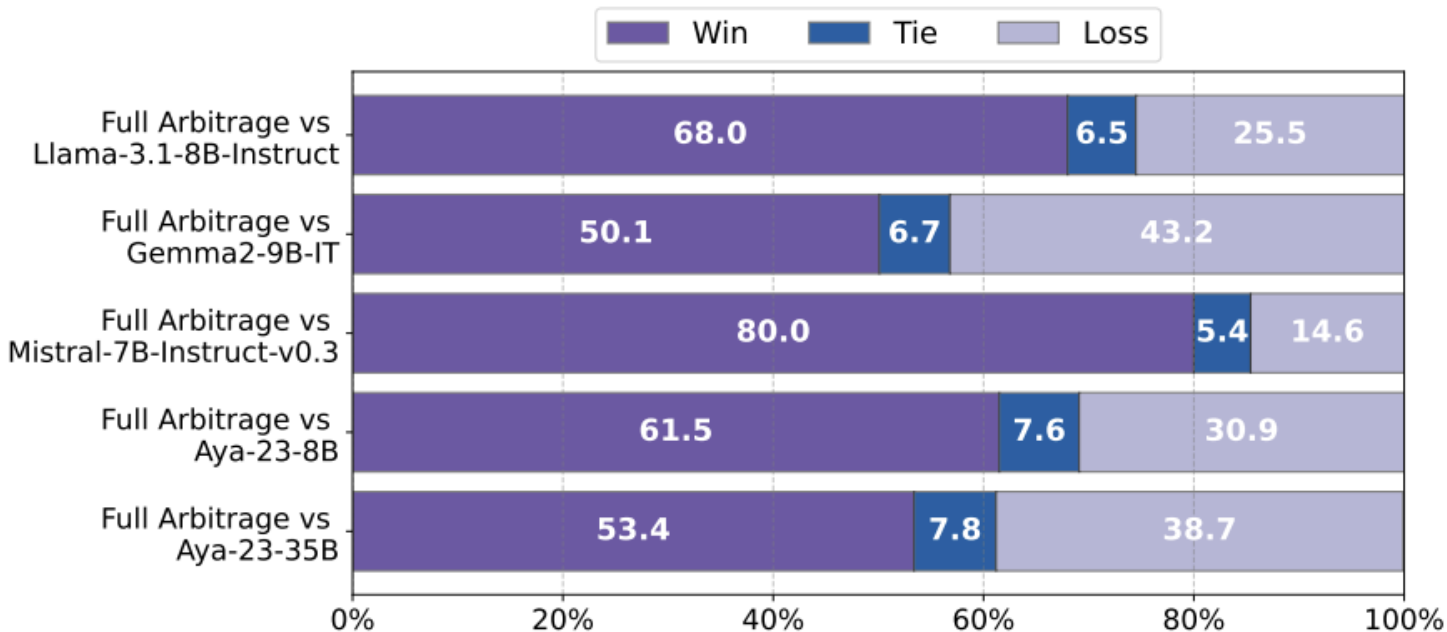
Multilingual Arbitrage

Compositions



Multilingual Arbitrage

Results



04

Revealing Better Signal

What if we we could learn a richer representation from existing data?

Why make learning implicit?

What else do we know ?

Prompt

Give me a one-liner law joke in Spanish.

Completion

¿Por qué los jueces no van al gimnasio?

Porque ya tienen suficiente poder en sus manos.

We actually know :

- The completion is in Spanish
- It's in the legal domain
- It's in the humour category
- It's 2 lines long

Defining the Taxonomy

Taxonomy [13 markers → 90 possible values]

```
<task>          ==> CodeGeneration, CodeTranslation, Summarization, ...  
<domain>        ==> Code, Math, Science, Legal, ...  
<language>       ==> English, French, German, Spanish, Hindi, ...  
<length_bucket> ==> concise, medium, long  
< ... >         ==> ...
```

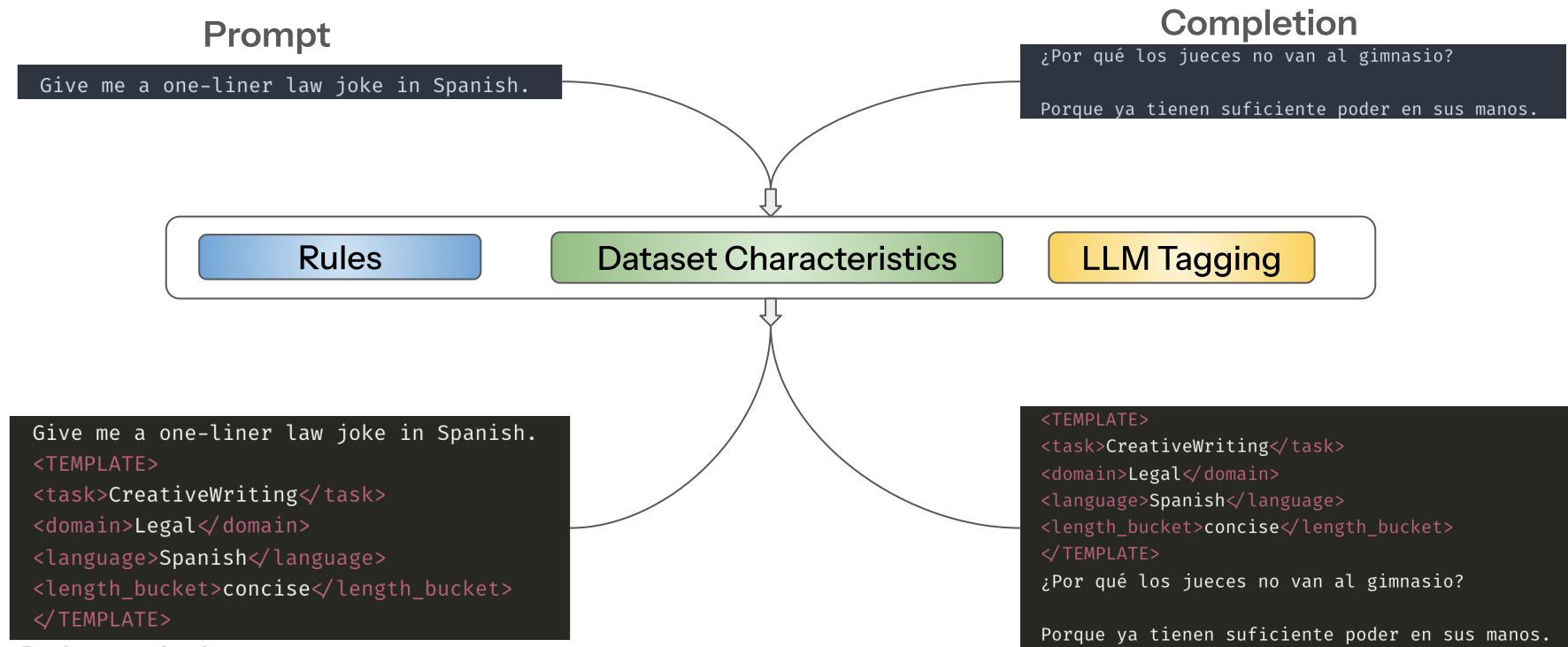
Annotation

Rules

Dataset Characteristics

LLM Tagging

Enriching the Dataset



Marker Dropout

Give me a one-liner law joke in Spanish.

```
<TEMPLATE>  
<domain>Legal</domain>  
</TEMPLATE>
```

```
<TEMPLATE>  
<task>CreativeWriting</task>  
<domain>Legal</domain>  
<language>Spanish</language>  
<length_bucket>concise</length_bucket>  
</TEMPLATE>
```

¿Por qué los jueces no van al gimnasio?

Porque ya tienen suficiente poder en sus manos.

Give me a one-liner law joke in Spanish.

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<TEMPLATE>  
<task>CreativeWriting</task>  
</TEMPLATE>
```

```
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<TEMPLATE>  
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</TEMPLATE>
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<task>CreativeWriting</task>  
<domain>Legal</domain>  
<language>Spanish</language>  
<length_bucket>concise</length_bucket>  
</TEMPLATE>
```

¿Por qué los jueces no van al gimnasio?

Porque ya tienen suficiente poder en sus manos.

Treasure-marked Model



How do I check for a palindrome in python?



<TEMPLATE>

<code_type>python</code_type>

<domain>Code</domain>

<source>AA</source>

<task>CodeGeneration</task>

</TEMPLATE>

A palindrome is a word, phrase, number, or other that reads the same forward and backward (ignoring punctuation and capitalization). To check if a given string is a palindrome, you can use the following code:

```
def is_palindrome(s):  
    """  
    Check if a string is a palindrome.  
    """
```



Can you give me a 3 sentence paragraph on the French Revolution ?



<TEMPLATE>

<domain>SocialSciences</domain>

<lang>English</lang>

<length_bucket>concise</length_bucket>

<length_paragraphs>1</length_paragraphs>

<length_sentences>3</length_sentences>

<length_tokens>81</length_tokens>

<quality>4.4</quality>

<quality_bucket>2</quality_bucket>

<source>CC</source>

<task>QuestionAnswering</task>

</TEMPLATE>

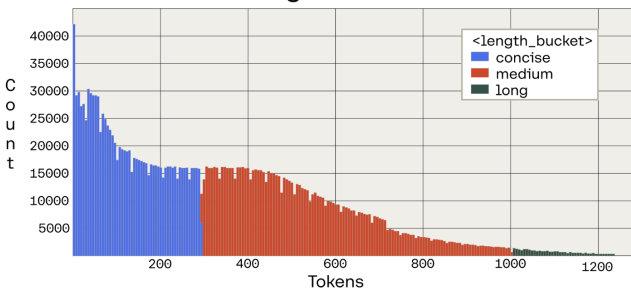
The French Revolution, which took place between 1789 and 1799, was a period of profound social and political upheaval in France. It marked the end of the French absolute monarchy and the creation of a republic, driven by ideals of liberty, equality, and fraternity. The revolution had a significant impact on European history, spreading the ideas of Enlightenment and leading to the Napoleonic Era and subsequent modernization of European political systems.

Treasure-marked Model

“Controlling” generations at inference

CONTROL AT
INFERENCE

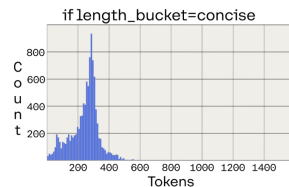
Training Distribution



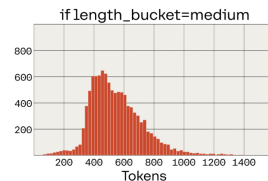
At inference time, **changing a single marker** can lead to **vastly different test-time distributions**

Distribution at Inference

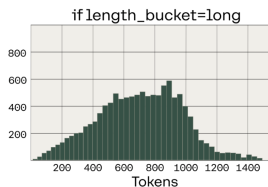
```
Tell me about the French Revolution  
<TEMPLATE>  
<length_bucket>concise</length_bucket>  
</TEMPLATE>
```



```
Tell me about the French Revolution  
<TEMPLATE>  
<length_bucket>medium</length_bucket>  
</TEMPLATE>
```



```
Tell me about the French Revolution  
<TEMPLATE>  
<length_bucket>long</length_bucket>  
</TEMPLATE>
```



Treasure Hunt

“Controlling” generations at inference



Reward Model Annotation Markers

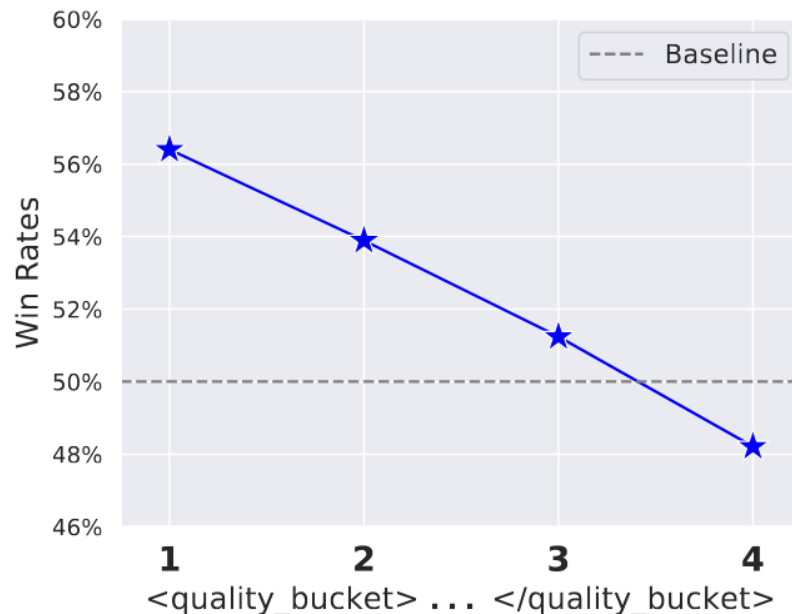
<quality_bucket> 1 ⇒ **Highest** Quality

<quality_bucket> 2 ⇒ ...

<quality_bucket> 3 ⇒ ...

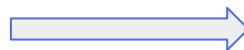
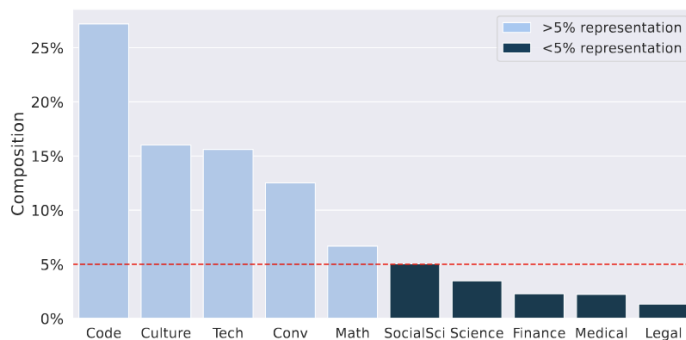
<quality_bucket> 4 ⇒ **Lowest** Quality

mArenaHard Win Rates against Baseline Model

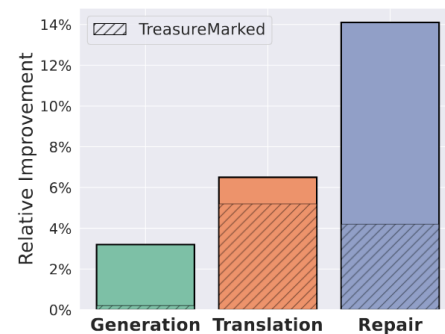
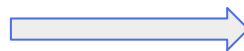
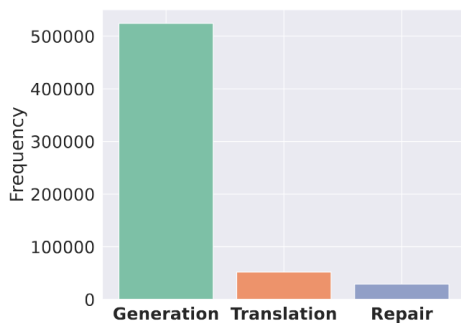
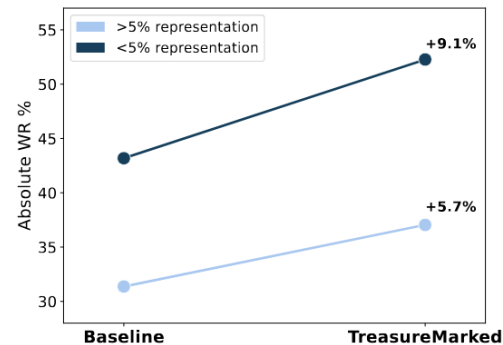


Results

Training Distribution

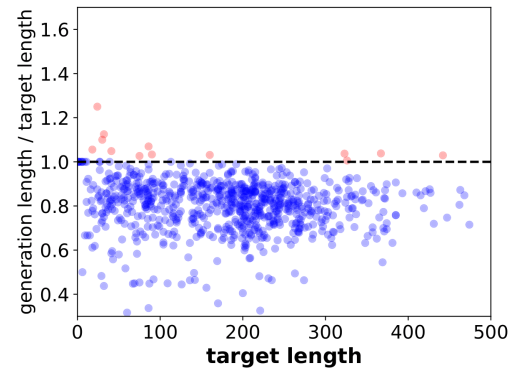
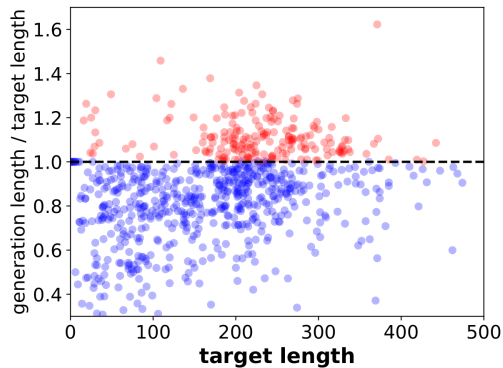
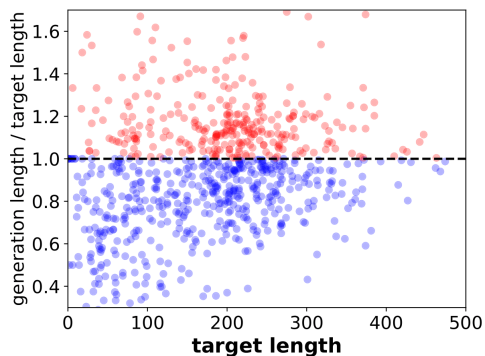


Improvement at Inference



Results

Alpaca Eval Length Instruct



	Baseline	TreasureMarked	+fixed
Violation Rate(%)	36.6 %	24.7 % ↓	1.3 % ↓
Win Rate(%)	14.4 %	19.5 % ↑	21.2 % ↑

05

Future Takeaways

Use Data as Leverage

- There are significant gains when focus shifts to **data**
- Simple optimizations in the data space are far more achievable than in the model space.
- Open Problems:
 - What are other avenues to generate “**better**” signal ?
 - How do we learn more “**explicit**” signals ?
 - What other “**assumptions**” do we hold about our data ?

Open for Questions!

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